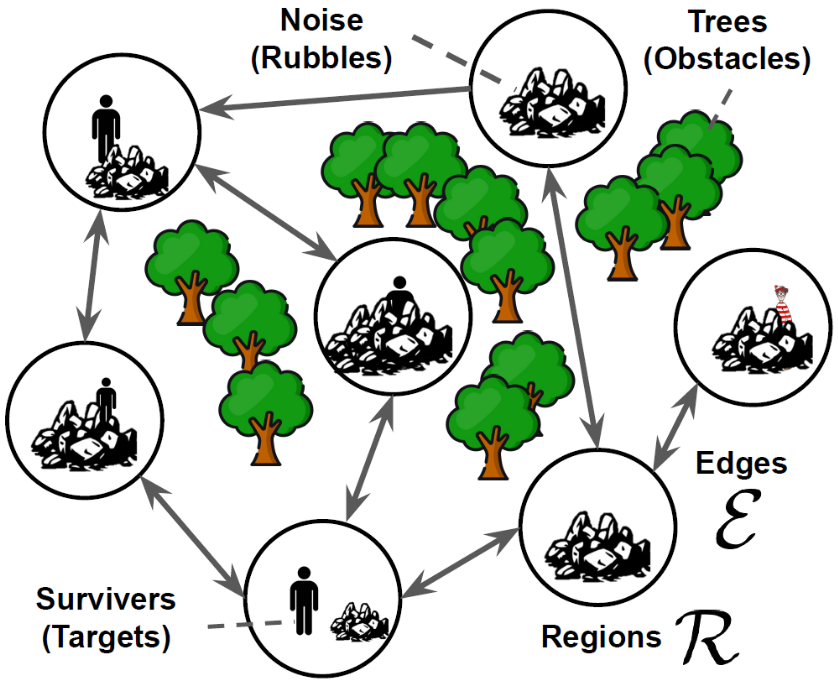
Percolation of information in an ergodic robotic network

Motivation

- **<u>Ergodicity</u>**: The total visitation frequency of each region is proportional to the **noise** of observation
- Swarm ergodicity requires all robots to have the same information about the regions
  - Previously done by central information processing
  - Decentralize by locally sharing collected information

How to maintain ergodicity



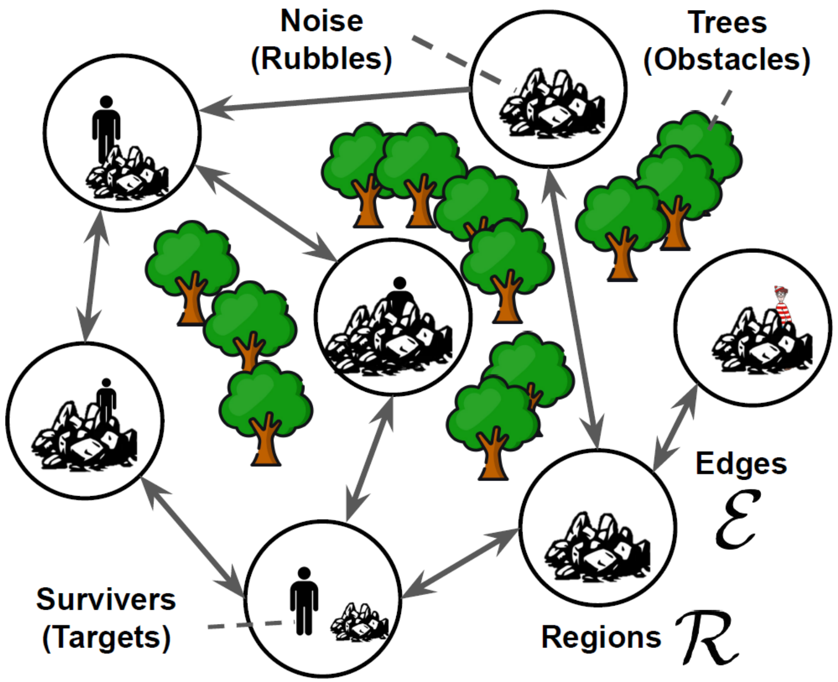
Recap

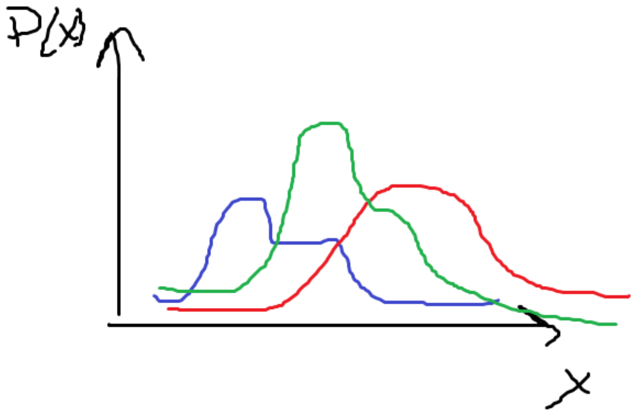
- Each region has a state to be estimated x_i
- The robots can make a noisy observation z_i when they transition to the region
- The estimation of x_i is done by recursive Baye's law

posterior Likelihood prior

$$P(x_i[k+1]) = \frac{P(z_{i,k}|x_i[k])P(x_i[k])}{P(z_{i,k})}$$

- If the robot starts with the same prior and knows all z_i, then all robot has the same posterior
- Mean and variance can be recovered from the posterior





Communication Protocol

- Samples can be reduced to ***sufficient statistic*** $\boldsymbol{\theta}$
  - e.g. mean μ and variance σ^2 for gaussian
  - Significantly reduce memory and communication
- Each robot keeps a list of $\boldsymbol{\theta}$ from the other robots
- When they communicate, they acquire the most recent $\boldsymbol{\theta}$ in their entire list from each other
- The robots communicate IFF they are in the same region

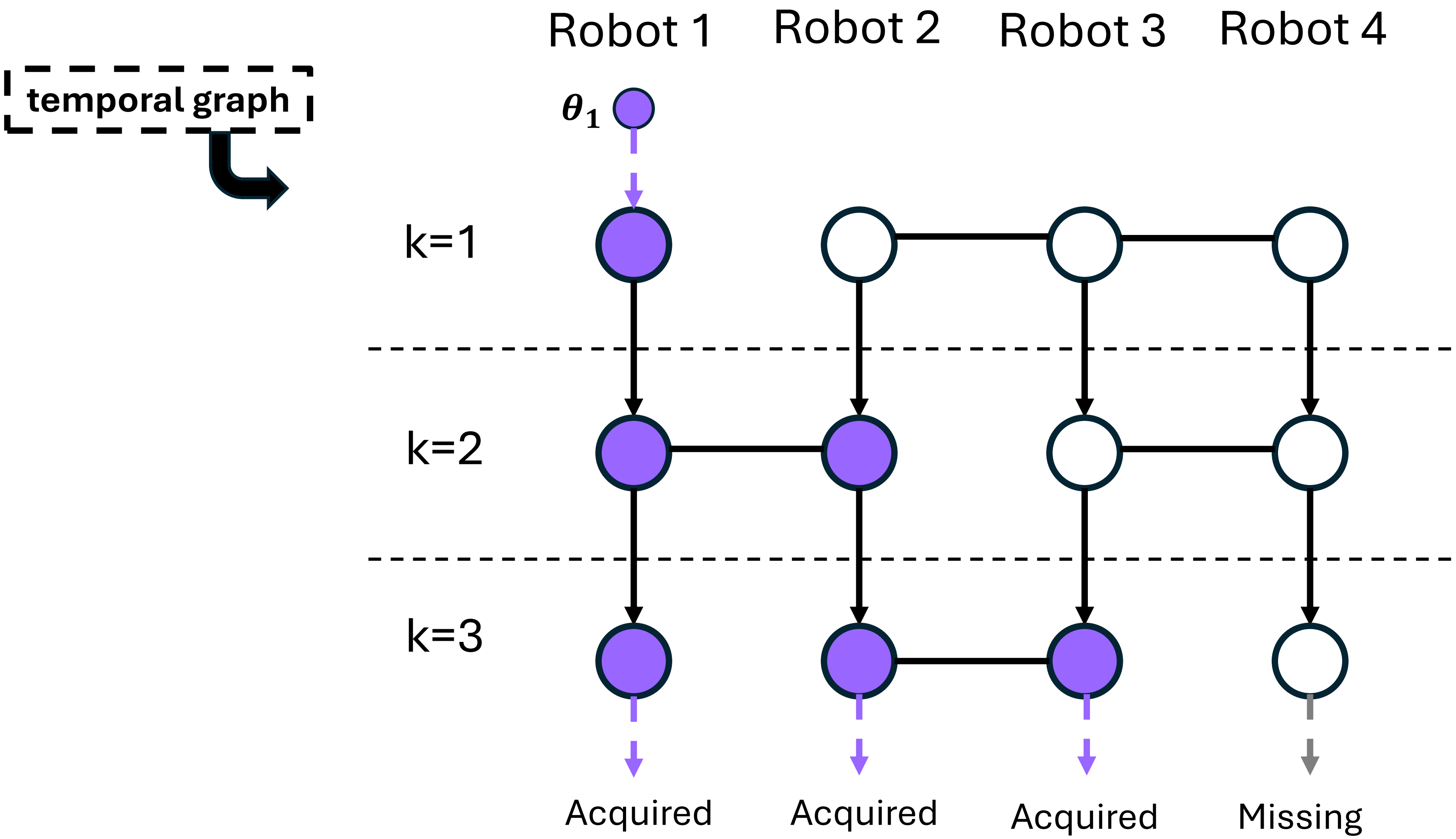
	Robot 1	**Robot 2**	**Robot 3**
Robot 1	$\boldsymbol{\theta}_{11}$	$\boldsymbol{\theta}_{12}$	$\boldsymbol{\theta}_{13}$
Robot 2	$\boldsymbol{\theta}_{21}$	$\boldsymbol{\theta}_{22}$	$\boldsymbol{\theta}_{23}$
Robot 3	$\boldsymbol{\theta}_{31}$	$\boldsymbol{\theta}_{32}$	$\boldsymbol{\theta}_{33}$

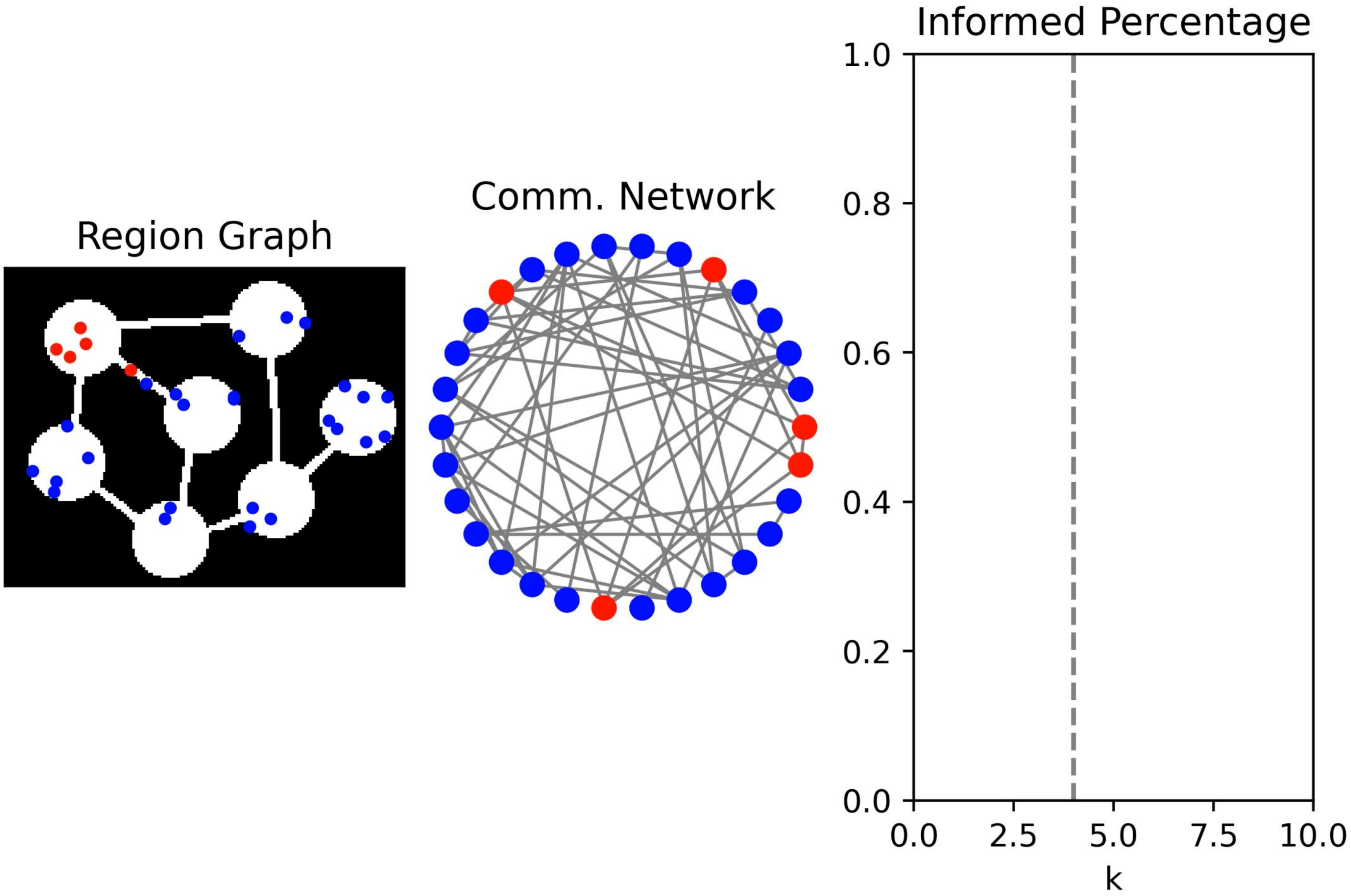
Problem Formulation

1. Assume every robot has a fixed initial piece of information
2. All robot has the same probability distribution of existing in the regions
3. How many time step can we expect all the initial information is shared to every robot?

Conjecture:
there is a sharp time step where the information is almost certainly shared across the robot network (as the number of robot tends to infinity)

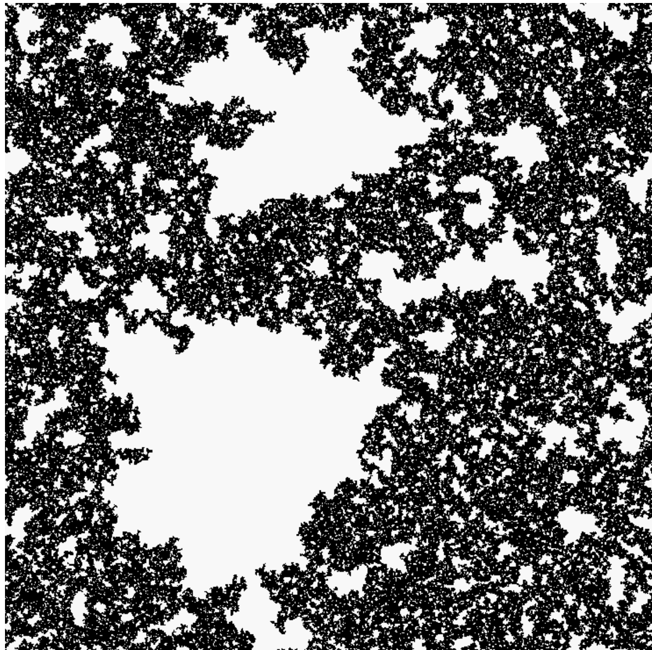
Pictorial Representation




Region Graph
Comm. Network
Informed Percentage
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
k

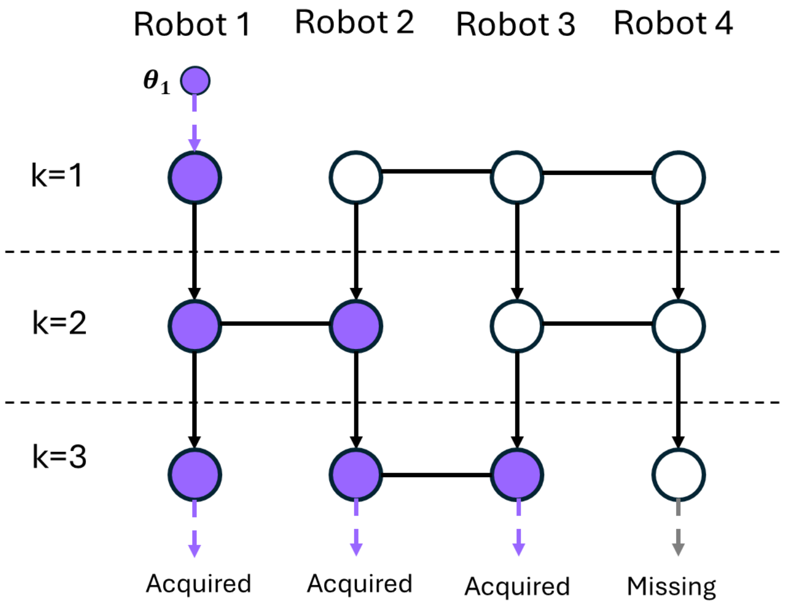
Percolation Theory

- Historically: a study of how fluid flow through porous material in statistical mechanic
  - e.g. water passing through coffee grounds
- In mathematics: A study of change of connectivity by adding random link in a graph
- Often shows **critical phenomenon**
  - A sudden change in behaviour when a parameter pass the **critical point** as it is being smoothly varied
  - e.g. ising model (magnets) , Erdős–Rényi model (random graph), phase change (boiling water)



Information Percolation

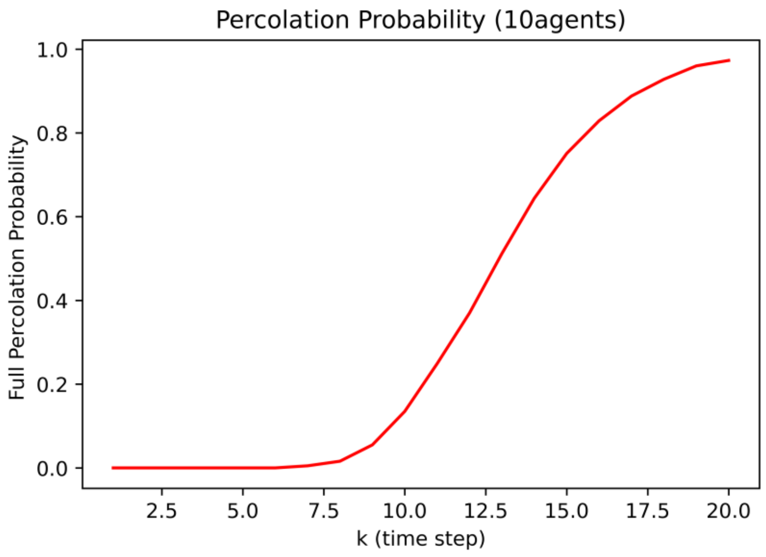
- Does the information show critical phenomenon
  - Is there a critical time window k_c where all information is guaranteed to be shared to all robot
  - i.e. there exist a path from all nodes to all nodes in the temporal graph
- How to model the critical time window
  - Given that the number of robots are known and the robots travel in the regions ergodically
  - 3 directions:
    - Direct modeling
    - Mean field approximation
    - Markov chain method

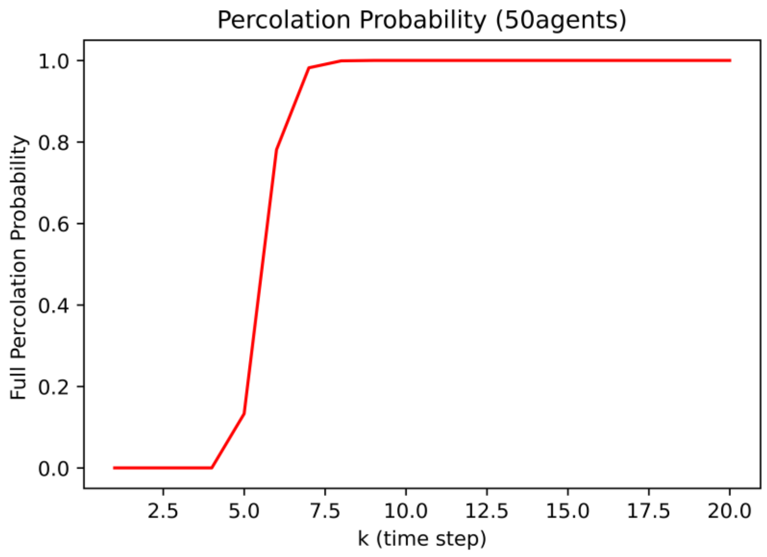


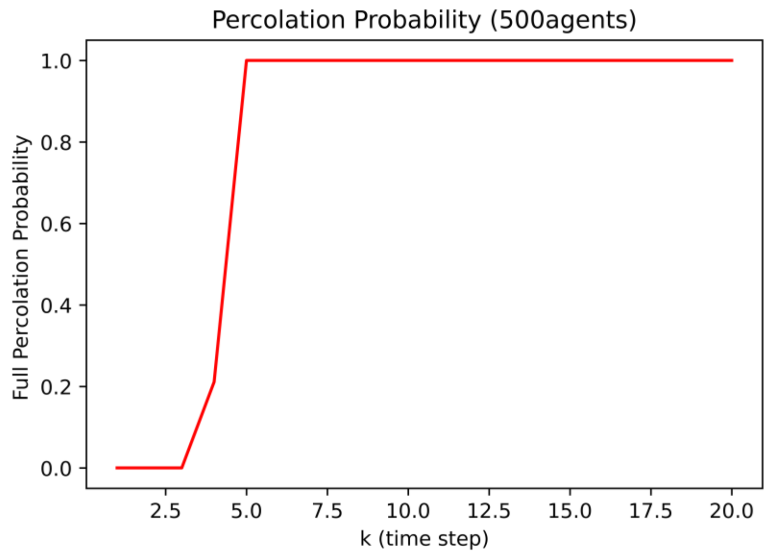
Simulated Result

Setting:
- 10 regions E-R graph
- Uniform distribution
- 1000 trials
- REMC transition law

- Key results:
  - Critical phenomenon exists
  - The more robots, the sharper the phase change is
  - The more robots, the smaller the critical time window is







Modeling – direct method

- From elementary probability properties
  - e.g.
    - $\Pr(2 \text{ robot meeting at time } k) = \sum p_i^2$
    - $\Pr(2 \text{ robot meeting at least once within horizon } K) = 1 - \left(1 - \sum p_i^2\right)^K$
- Ignore the effect of transitions
  - i.e. assume to robot can go to any region independent of the current region according to the spatial distribution
  - “steady state approximation”

List of probability properties

$$\neg(A\&B) = (\neg A)\,||(\neg B)$$
$$\neg(A||B) = (\neg A)\&(\neg B)$$

De Morgan's laws

$$P(A \,\&\, B) = P(A) \ if \ A \subset B$$
$$P(A \,||\, B) = P(B) \ if \ A \subset B$$

$$P(A|B) = \frac{P(A \,\&\, B)}{P(B)}$$

Bayes law

$$P(A) + P(\neg A) = 1$$
$$P(A|B) + P(\neg A|B) = 1$$

complement

$$P(A \,\&\, B) = P(A)P(B|A) = P(B)P(A|B)$$
$$P(A \,||\, B) = P(A) + P(B) \text{-} P(A \,\&\, B)$$
$$P(A \,\&\, B) = P(A) + P(B) \text{-} P(A \,||\, B)$$
$$P((\neg A)\&B) = P(B) \text{ -} P(A\&B)$$
$$P(A||B) = P(A) + P(\neg A\&B)$$

$$P(A||B||C) = P(A) + P(B) + P(C) \text{-} P(AB) \text{-} P(AC) \text{-} P(BC) + P(A\&B\&C)$$
$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum P(A_i) \text{ - } \sum P(A_i\&A_j) + \sum P\big(A_i\&A_j\&A_k\big) + \cdots + (\text{-}1)^{n\text{-}1} \bigcap_{i=1}^{n} A_i$$

Direct method – 3 robot system

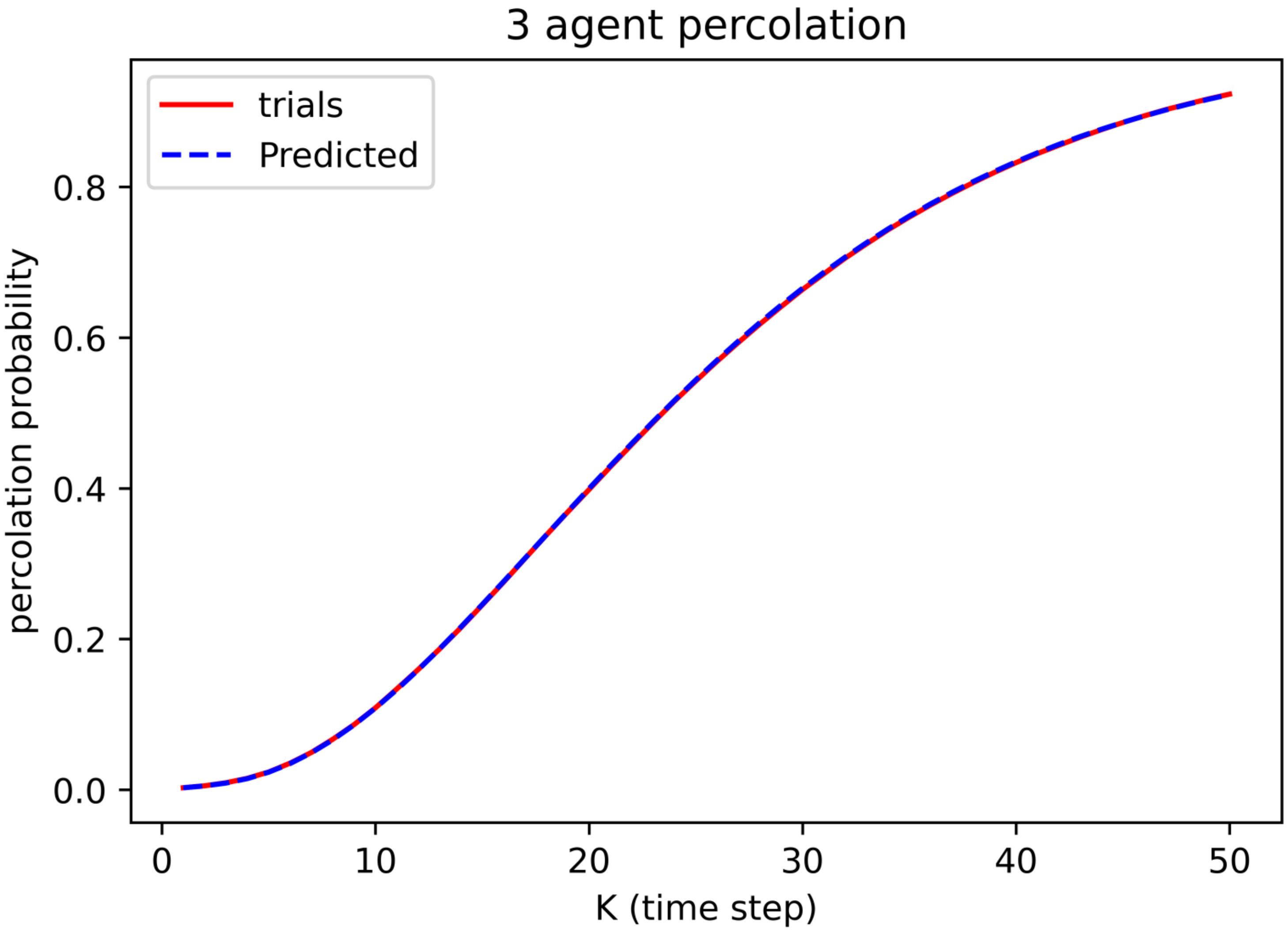
1. Find the condition for single source percolation (just percolate information of robot 1)
   - (1 meet 2 at least once) and (2 meet 3 at least once) or (1 meet 3) etc.
   - Number of conditions = number of ***spanning tree*** rooting at robot 1

$$\Pr(\text{percolate } \theta_1) = \Pr(((2-3) \leftarrow (1-2)) \text{ or } (1-3))\text{and}(((2-3) \leftarrow (1-3)) \text{ or } (1-2))$$

2. Combine the conditions of all sources by intersection

$$\Pr\begin{bmatrix}\text{Percolate } \theta_1 \\ \text{and} \\ \text{Percolate } \theta_2 \\ \text{and} \\ \text{Percolate } \theta_3\end{bmatrix} = \Pr\left(\begin{bmatrix}((2-3) \leftarrow (1-3)) \text{ and } \big((1-3) \leftarrow (2-3)\big) \\ \text{or} \\ ((2-3) \leftarrow (1-2)) \text{ and}\big((1-2) \leftarrow (2-3)\big) \\ \text{or} \\ ((1-3) \leftarrow (1-2)) \text{ and}\big((1-2) \leftarrow (1-3)\big) \\ \text{or} \\ (1-2)\text{and}(1-3)\text{and}(2-3)\end{bmatrix}\right)$$

3 - robot results



Direct Method Problem

- Number of conditions = number of ***spanning tree*** in the robot network
  - The number of spanning tree is provided by the **Cayley's Formula**

$$N^{N-2}$$

- For 4 robots, full percolation has 16^4 conditions, impossible to derive by hand

Mean Field Approximation

- Popular method for handling statistic problem with large amount of variables
  - e.g. Ising model, Navier-Stokes, **pandemic modelling**
- Treat the swarm as continuous variables and approximate the dynamic using expected value
  - E.g. it is possible to have 1.53 robot in a region

Mean Field Approximation - Formulation

- <u>Only consider single source for now</u>
- Define the proportion of robot that has the information as P_I
  - The rest is $1 - P_I$
- At any time if a region has at least 1 informed robot, the rest will also become informed, the expected increase of informed robot is

$$\Delta P_I = \sum_i \underbrace{(1 - (1 - p_i)^{P_I N})}_{\text{Probability of at least one informed robot in the region}} \underbrace{(1 - P_I) p_i}_{\text{Expected number of uninformed robot in the region}}$$

Solve the difference equation

$$P_I[k+1] = P_I[k] + \sum_i \left(1 - (1 - p_i)^{P_I[k] N}\right)(1 - P_I[k]) p_i$$

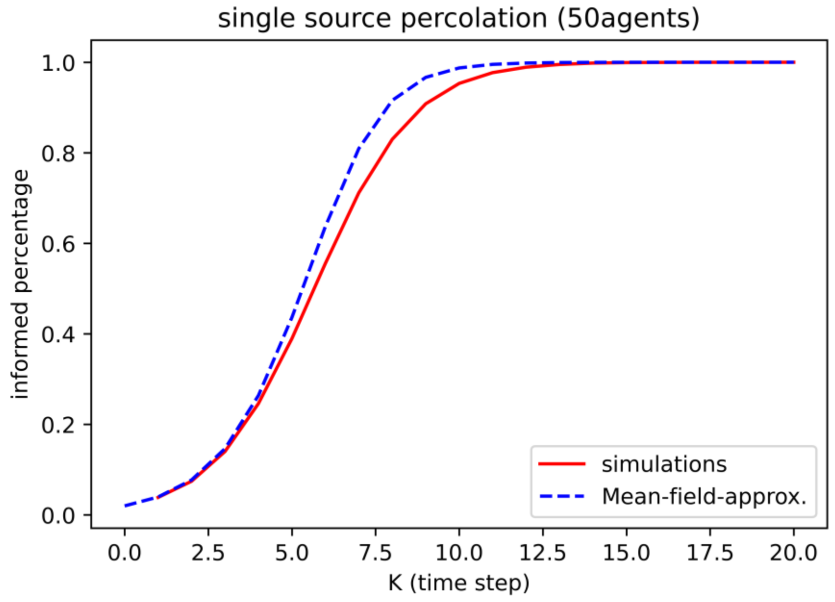
$$P_I[0] = \frac{1}{N}$$

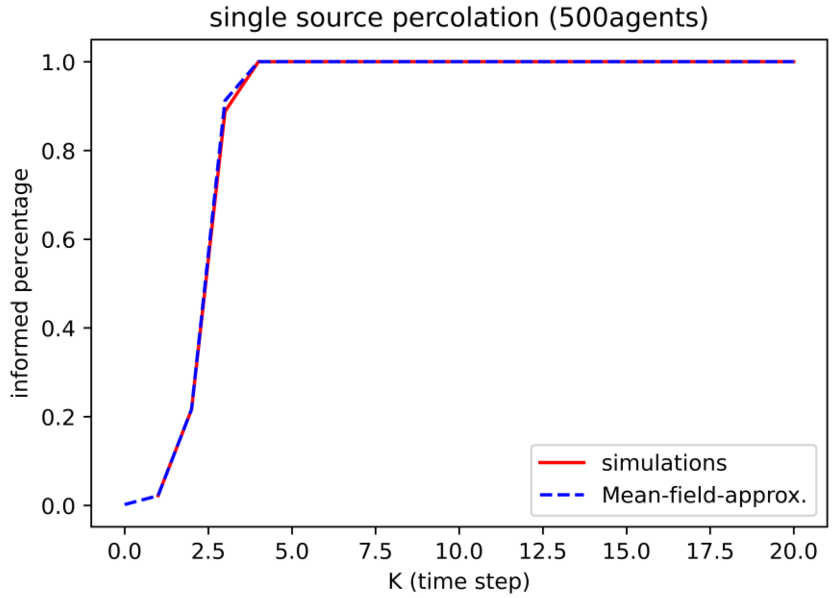
Only 1 informed robot out of N initially

Mean Field Approx – Result

<u>**IMPORTANT: y-axis is different from previous plots**</u>

- Only good for a lot of robots
  - underestimate with few robots
- Hard to translate to critical percolation
  - The difference equation only approach 1
  - $\Rightarrow$ Full percolation requires infinite time in MFA
- Don't know how to handle multi-source percolation yet

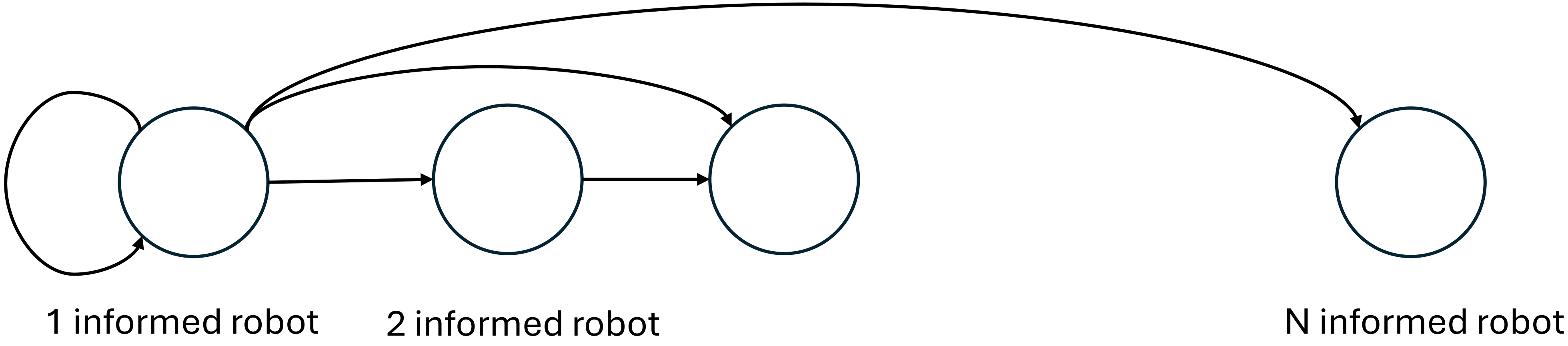




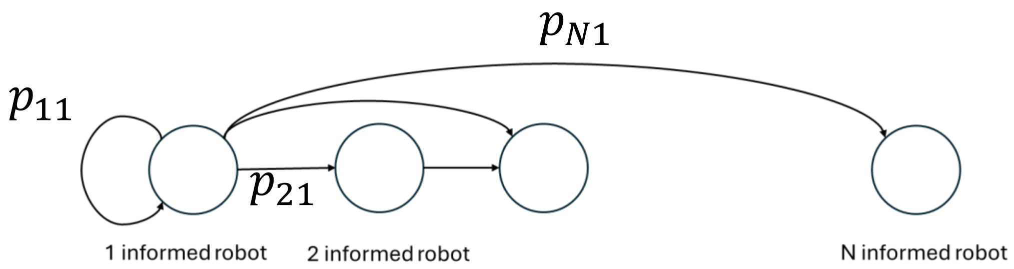
Markov Chain Formulation

IMPORTANT: this Markov chain is not related to the ergodic Markov chain

- Consider the number of informed robot as a state
- Every time step, after the robot moved, there is a chance of the numer of informed robots increasing (transition to subsequent state) or stays constant (self loop). But never decrease (go back to previous states)
- Eventually the chain terminates at all robots being informed
  - This is called an absorbing chain



Expected Absorption Time



- The corresponding stochastic matrix is

$$\boldsymbol{Q}$$

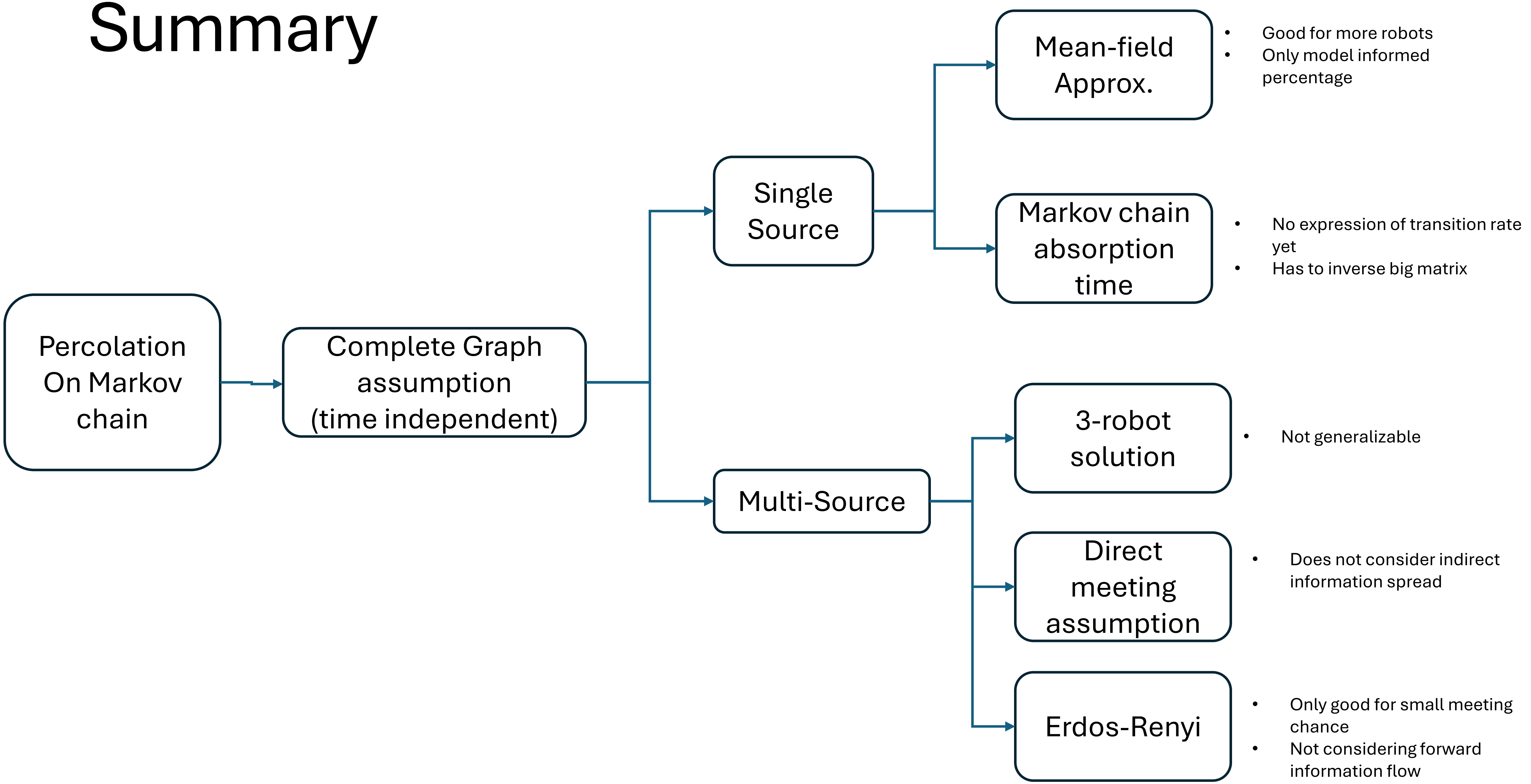
$$M = \begin{bmatrix} p_{11} & 0 & 0 & \cdots & 0 \\ p_{21} & p_{22} & 0 & \cdots & \vdots \\ \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & p_{nr} & \ddots & 0 \\ p_{N1} & p_{N2} & \cdots & & 1 \end{bmatrix}$$

- The expected absorption time is given by

$$t = \mathbf{1}^T (I - Q)^{-1} [1 \quad 0 \quad 0 \quad \cdots]^T$$

- Need to find the expression for p_{nr}(probability of informed robot increased from r to n in 1 step)
  - significantly easier than direct method (but still hard)
    - Mean field approx?
- Not scalable with a lot of robots (inverting (N-1)X(N-1) matrix)
- Need to formulate as multi-source percolation

Summary

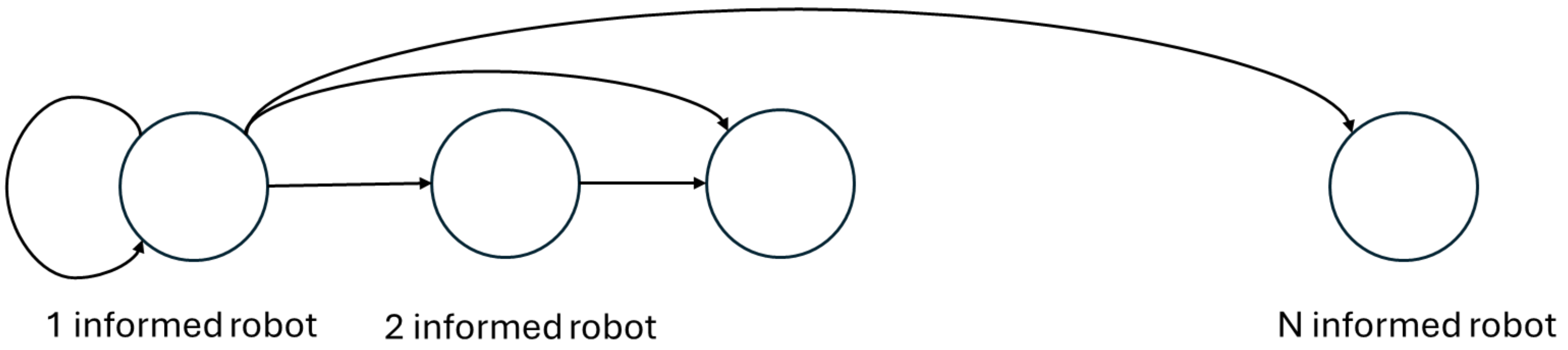


Percolation On Markov chain
Complete Graph assumption (time independent)
Single Source
Multi-Source
Mean-field Approx.
Markov chain absorption time
3-robot solution
Direct meeting assumption
Erdos-Renyi
• Good for more robots
• Only model informed percentage
• No expression of transition rate yet
• Has to inverse big matrix
• Not generalizable
• Does not consider indirect information spread
• Only good for small meeting chance
• Not considering forward information flow

1/29/26 Updates

Recep

- Single source percolation can be estimated with first hit model if the transition probability of the Markov model can be found
  - E.g. If 2 robot is informed, what is the probability that 5 will be informed in the next step



First Hitting Time Method (fully connected)

<u>Pairwise approximation</u>

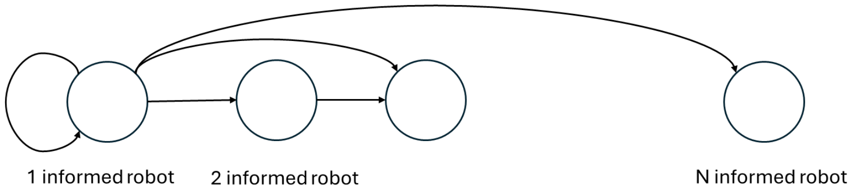
(accurate if meeting chance is not too high)

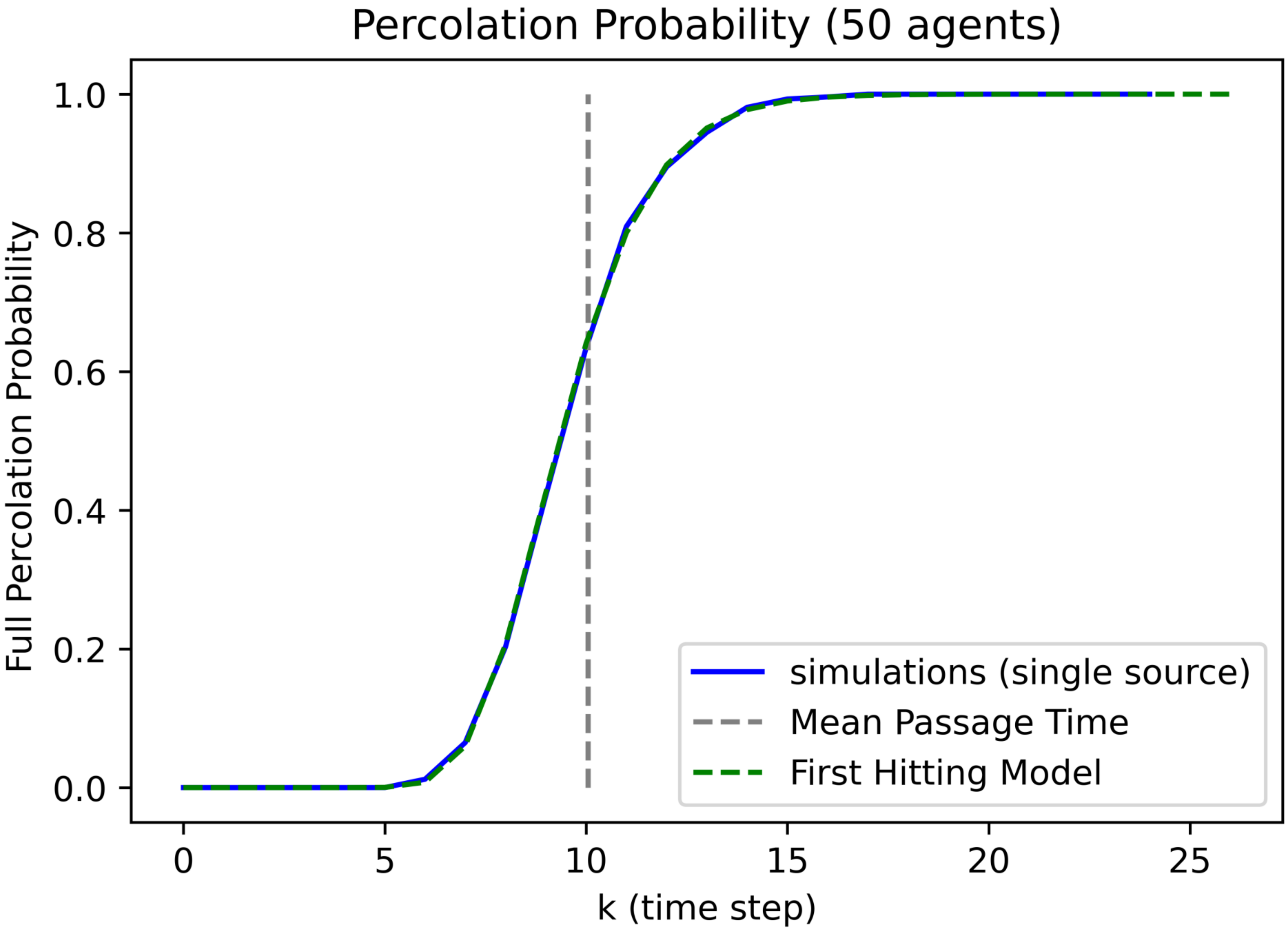
$$p_{nr} = \Pr(r \text{ newly informed}|n \text{ informed}) = \underbrace{C_r^{N-n}}_{\text{How many way to get r uninformed robots}} \underbrace{(1-(1-\overbrace{S_2}^{\text{Pairwise meeting chance (fully connected)}})^n)^r}_{\text{r robots meet with at least 1 informed robots}} \underbrace{\left((1-S_2)^n\right)^{N-n-r}}_{\text{Rest of the uninformed robot does not meet with any informed robot}}$$

$$\Pr(N\ informed\ at\ k) = 1 - \underbrace{\mathbf{1}^T Q^k \overbrace{[1 \quad 0 \quad 0 \quad \cdots]^T}^{\text{Initial distribution}}}_{\text{Probability of not absorbed in k step}}$$

Q

$$M = \begin{bmatrix} p_{11} & 0 & 0 & \cdots & 0 \\ p_{21} & p_{22} & 0 & \cdots & \vdots \\ \vdots & \vdots & \ddots & & \vdots \\ \vdots & \vdots & p_{nr} & \ddots & 0 \\ p_{N1} & p_{N2} & \cdots & & 1 \end{bmatrix}$$



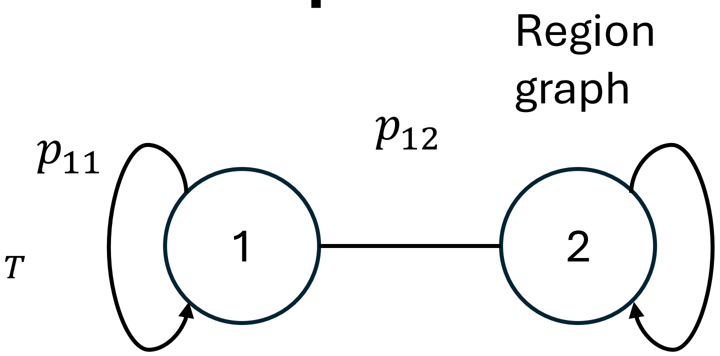

Percolation Probability (50 agents)
Full Percolation Probability
k (time step)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
simulations (single source)
Mean Passage Time
First Hitting Model

Pairwise Meeting Chance on Region Graph

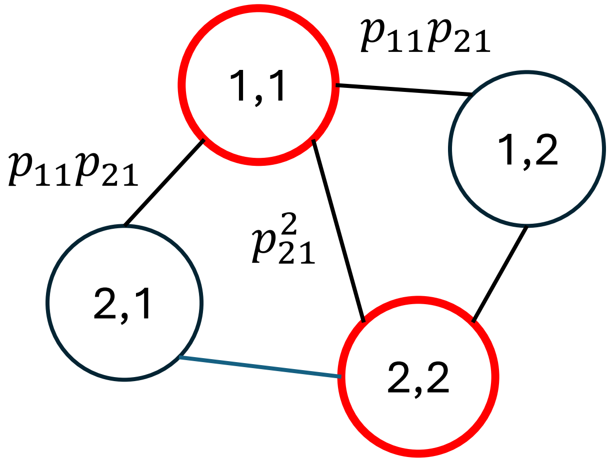
- Consider 2 agents on a region graph with a given transition matrix
- Create new state space with the cartesian product
- Find initial state in combined state space
- Remove "collision states"

Initial distribution

$$\rho_0 = [\rho_{0,1} \quad \rho_{0,2}]^T$$



Cartesian product



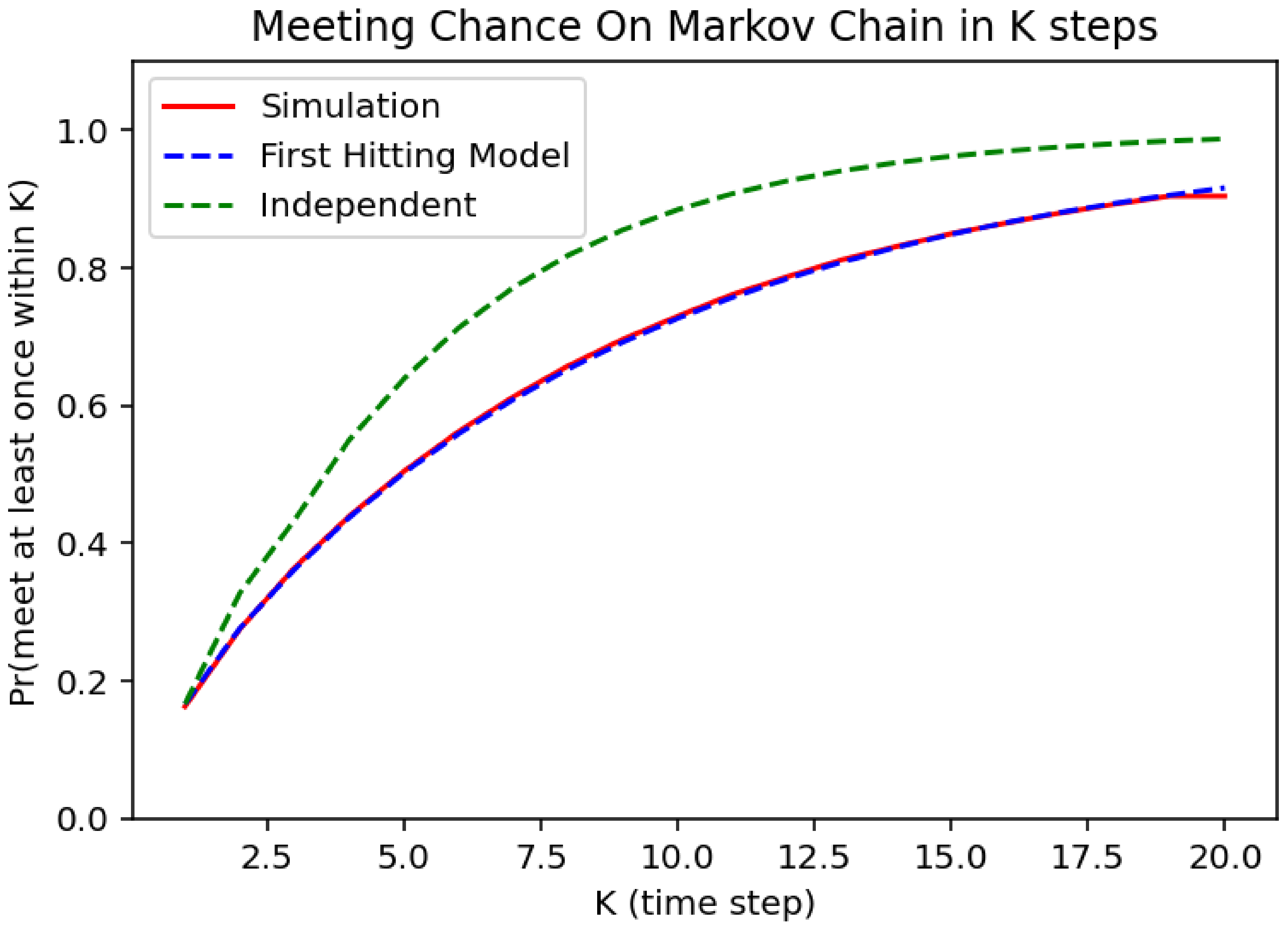
$$\Pr(\mathit{meet\ at\ least\ once\ within\ k}) = 1 - \mathbf{1}^T Q^k q_o$$

$$M = \begin{array}{c} \begin{array}{cccc} (1,1) & (1,2) & (2,1) & (2,2) \end{array} \\ \begin{bmatrix} p_{11}^2 & p_{11}p_{12} & p_{12}p_{11} & p_{12}^2 \\ p_{11}p_{21} & p_{11}p_{22} & p_{12}p_{21} & p_{12}p_{22} \\ p_{21}p_{11} & p_{21}p_{12} & p_{22}p_{11} & p_{22}p_{12} \\ p_{21}^2 & p_{21}p_{22} & p_{12}p_{11} & p_{22}^2 \end{bmatrix} \end{array}$$

Q

$$m_0 = [\rho_{0,1}^2 \quad \rho_{0,1}\rho_{0,2} \quad \rho_{0,2}\rho_{0,1} \quad \rho_{0,2}^2]^T$$

q_0



Meeting Chance On Markov Chain in K steps
Simulation
First Hitting Model
Independent
Pr(meet at least once within K)
K (time step)
0.0
0.2
0.4
0.6
0.8
1.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

Conditional Pairwise Meeting Chance

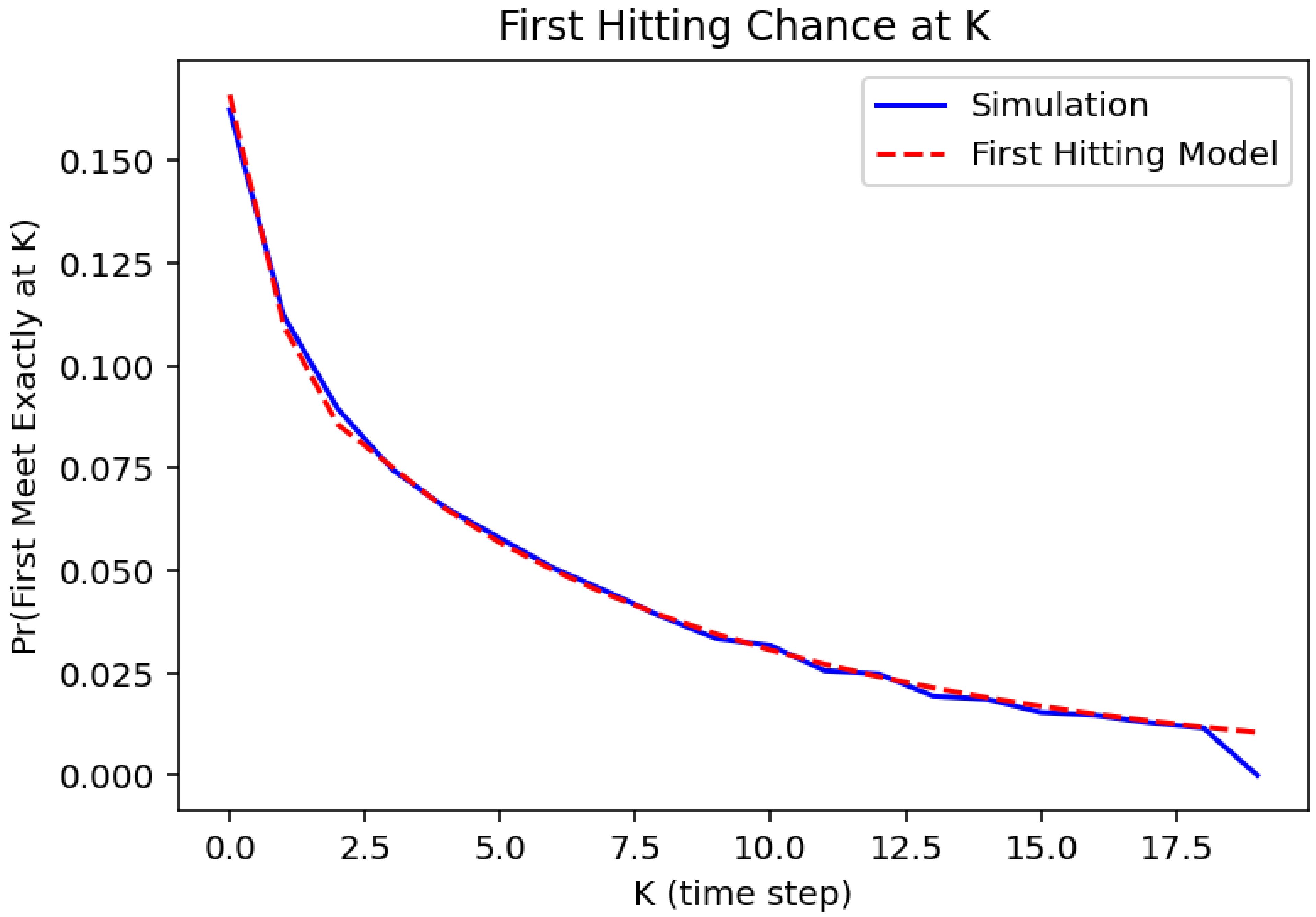
- We need the conditional meeting chance
  - Given that the robot never met an informed robot, what is the chance of meeting an informed robot in the next step

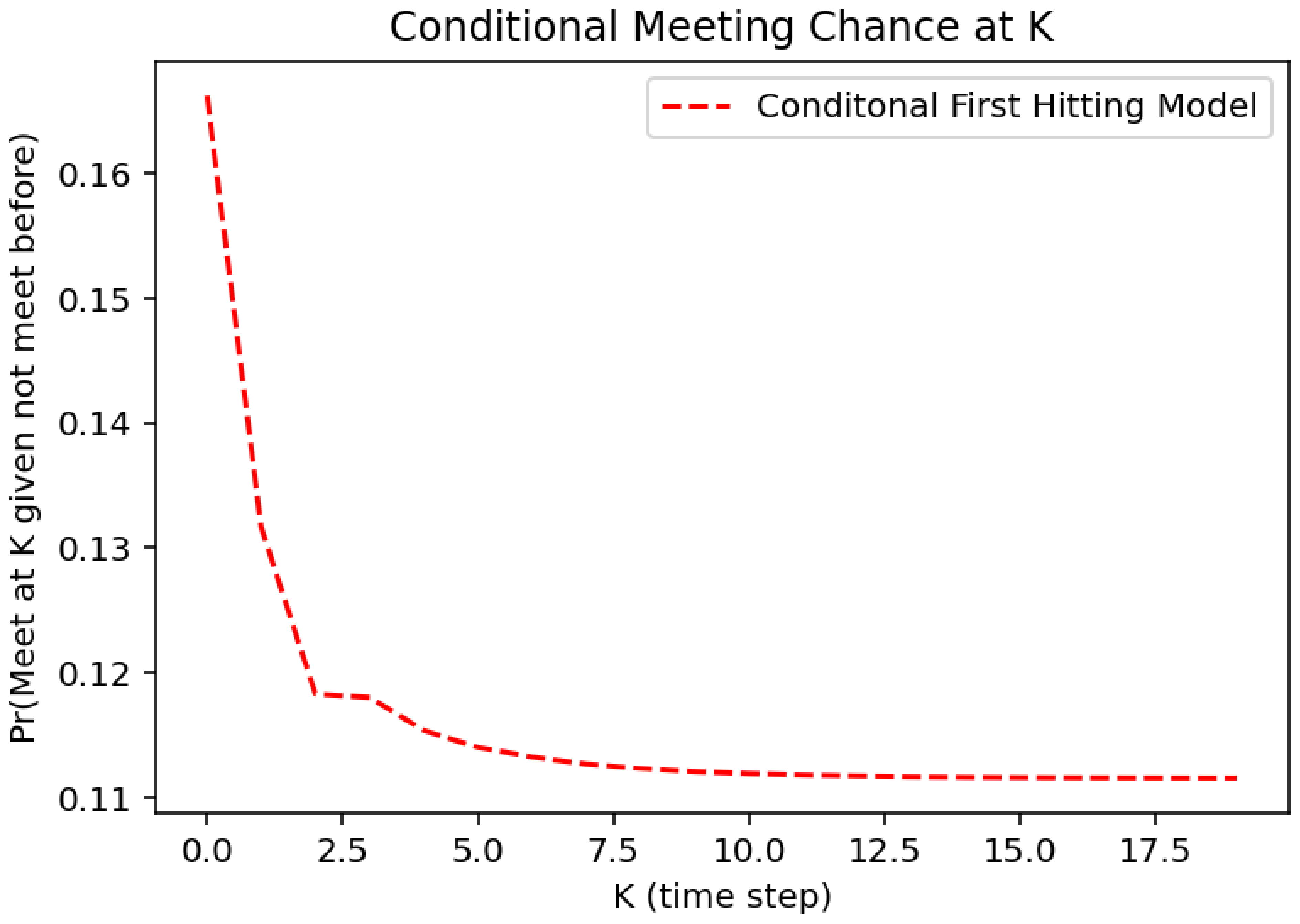
$$\Pr(first\ meet\ at\ k) = \overset{\text{1s at collision states}}{[1 \quad 0 \quad 0 \quad 1 \quad \cdots]}\underbrace{PQ^{k-1}q_0}_{\substack{\text{Space distribution never met} \\ \text{from 0 to k-1}}}$$

$$\Pr(meet\ at\ K\ |never\ met\ before) = \frac{\Pr(first\ meet\ at\ K)}{1 - \sum_1^{K-1} \Pr(first\ meet\ at\ k)}$$

(Probably can be simplified)

- Interesting effect: the longer K is, the lower this probability is, but approach a fix value for very big K
- Interpretation: if they haven't meet in a long time, they probably are initially very far away in the region graph


First Hitting Chance at K
Simulation
First Hitting Model
Pr(First Meet Exactly at K)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
K (time step)


Conditional Meeting Chance at K
Conditonal First Hitting Model
Pr(Meet at K given not meet before)
0.16
0.15
0.14
0.13
0.12
0.11
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
K (time step)

Combine Meeting chance and percolation

Fully connected percolation transition probability

$$p_{nr} = \Pr(r \text{ newly informed} | n \text{ informed}) = C_r^{N-n}(1-(1-S_2)^n)^r\left((1-S_2)^n\right)^{N-n-r}$$

Now its time varying

$$p_{nr}[k] = \Pr(r \text{ newly informed} | n \text{ informed}) = C_r^{N-n}(1-(1-S_2)^n)^r\left((1-S_2)^n\right)^{N-n-r}$$

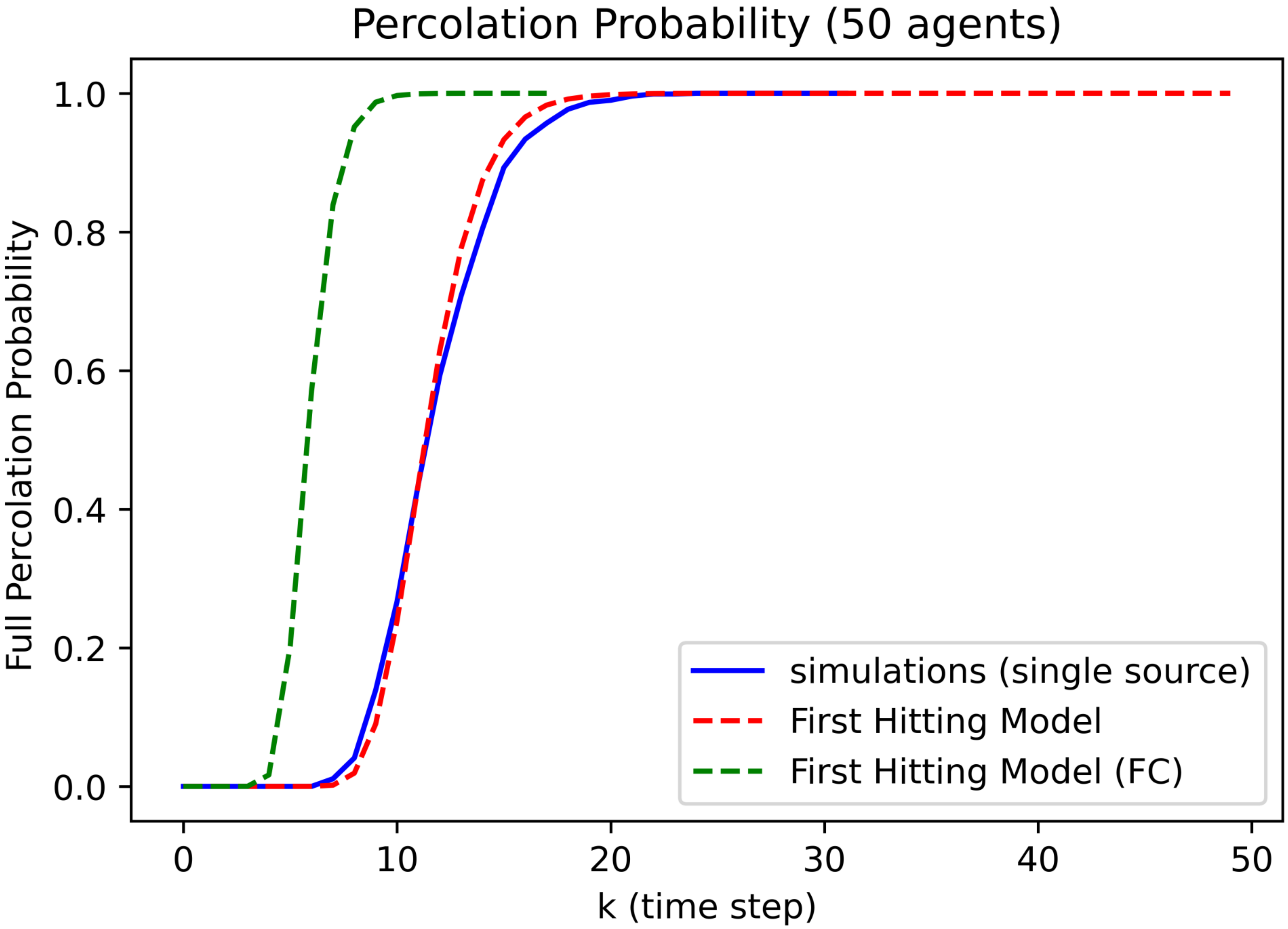
Replace S_2 with the conditional probability

$$\Pr(meet\ at\ K\ | never\ meet\ before) = \frac{\Pr(first\ meet\ at\ K)}{1-\sum_{k=0}^{K-1}\Pr(first\ meet\ at\ k)}$$

Time varying first hit model

$\Pr(meet\ at\ least\ once\ within\ k) = 1 - \mathbf{1}^T Q^k q_o$ Original first hit model

$\Pr(meet\ at\ least\ once\ within\ K) = 1 - \mathbf{1}^T \left(\prod_{k=0}^{K} Q[k] \right) q_o$ Time varying first hit model



Percolation Probability (50 agents)
Full Percolation Probability
k (time step)
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
simulations (single source)
First Hitting Model
First Hitting Model (FC)

2/28/26

Graph-MFA

$$I[k+1] = P(I[k] + \Delta I[k])$$

$$U[k+1] = P(U[k] - \Delta I[k])$$

$$\Delta I_i[k] = U_i[k]\left(1 - \left(1 - \frac{I_i[k]}{\sum I_j[k]}\right)^{\sum I_j[k]}\right)$$

$$I = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ \vdots \\ I_n \end{bmatrix}$$

Number of informed robots in each region

$$U = \begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ \vdots \\ U_n \end{bmatrix}$$

Number of uninformed robots in each region

$$1^{\mathsf{T}}(I + U) = N$$

Problem: probability is not independent in time

Informed probability

$$p_{nr} = \Pr(r \text{ newly informed}|n \text{ informed}) = C_r^{N-n}(1-(1-S_2)^n)^r\left((1-S_2)^n\right)^{N-n-r}$$

$$p_{nr} = \Pr(r \text{ newly informed}|n \text{ informed}) = C_r^{N-n}(\Pr(\text{informed at } k))^r(1-\Pr(\text{informed at } k))^{N-n-r}$$

- Need $\Pr(\text{informed at } k|\text{ not informed for } k-1, I[k])$

$$\Pr(\text{informed at } k|\text{ not informed for } k-1, I[k]) = \frac{P(\text{first informed at } k \mid I[k])}{P(\text{not informed for } k-1 \mid I[k])}$$

i.e. It is more likely to not be informed if haven't been inform for a long time

03/09/2026

Approximation?

$$\Pr(\text{First Informed } at\ k) = \sum_{i=0} \rho_i[k](1 - \left(1 - \frac{I_i[k]}{\sum_j I_j}\right)^{\sum_j I_j})$$

Autocorrelation $\approx \lambda_2$ If $\rho[0] = \bar{\rho}$

$$\Pr(\text{First Informed } at\ k | \text{ not informed before } k) \leq \frac{1}{1+\lambda_2} \Pr(\text{First Informed } at\ k)$$

$$\rho \triangleq \begin{bmatrix}\rho_1\\ \rho_2\\ \vdots\end{bmatrix} \quad P \triangleq \begin{bmatrix}P_{11} & P_{12} & \cdots\\ P_{21} & P_{22} & \cdots\\ \vdots & \vdots & \ddots\end{bmatrix}$$

$$\rho^2 \triangleq \begin{bmatrix}\rho_1^2\\ \rho_2^2\\ \vdots\\ \rho_1\rho_2\\ \rho_1\rho_3\\ \vdots\end{bmatrix} = \begin{bmatrix}\rho_{meet}^2\\ \rho_{not\ meet}^2\end{bmatrix}$$

Combined distribution

$$\begin{array}{ccccccc} (1,1) & (2,2) & & (1,2) & (1,3) & \end{array}$$

$$P_{\mathcal{G}^2} \triangleq \left[\begin{array}{ccc|ccc} P_{11}^2 & P_{12}^2 & \cdots & P_{11}P_{12} & P_{11}P_{13} & \cdots\\ P_{21}^2 & P_{22}^2 & \cdots & P_{21}P_{22} & P_{21}P_{23} & \cdots\\ \vdots & \vdots & \ddots & \vdots & \vdots & \cdots\\ \hline P_{11}P_{21} & P_{12}P_{22} & \cdots & P_{11}P_{22} & P_{11}P_{23} & \cdots\\ P_{11}P_{31} & P_{12}P_{32} & \cdots & P_{11}P_{32} & P_{11}P_{33} & \cdots\\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}\right] = \begin{bmatrix}A & P_{meet}\\ C & P_{\neg\ meet}\end{bmatrix}$$

Combined transition

$$\Pr(\text{Meeting at K}) = [\mathbf{1}_n^T \quad \mathbf{0}]\left(P_{\mathcal{G}^2}\right)^K \rho^2[0] = (\rho[K])^T \rho[K]$$

$$\Pr(\text{Not Meeting within K}) = [\mathbf{0}_n \quad \mathbf{1}^T]\begin{bmatrix}0 & 0\\ 0 & P_{\neg meet}\end{bmatrix}^K \begin{bmatrix}0\\ \rho^2_{\neg\ meet}[0]\end{bmatrix}$$

$$\Pr(\text{First Meeting at K}) = [\mathbf{1}_n^T \quad \mathbf{0}]\left(P_{\mathcal{G}^2}\right)\begin{bmatrix}0 & 0\\ 0 & P_{\neg meet}\end{bmatrix}^{K-1} \begin{bmatrix}0\\ \rho^2_{\neg meet}[0]\end{bmatrix}$$

$$\Pr(\text{Meeting at K}|\ \text{not met before } K) = \frac{\Pr(\text{First Meeting at K})}{\Pr(\text{Not Meeting within K} - 1)} = \frac{[\mathbf{1}_n^T \quad \mathbf{0}]\left(P_{\mathcal{G}^2}\right)\begin{bmatrix}0 & 0\\ 0 & P_{\neg meet}\end{bmatrix}^{K-1}\begin{bmatrix}0\\ \rho^2_{\neg meet}[0]\end{bmatrix}}{[\mathbf{0}_n \quad \mathbf{1}^T]\begin{bmatrix}0 & 0\\ 0 & P_{\neg meet}\end{bmatrix}^{K-1}\begin{bmatrix}0\\ \rho^2_{\neg meet}[0]\end{bmatrix}} = \frac{\mathbf{1}_n^T P_{meet} P_{\neg meet}^{K-1} \rho^2_{\neg meet}[0]}{\mathbf{1}_n^T P_{\neg meet}^{K-1} \rho^2_{\neg meet}[0]}$$

$$\Pr(\text{First Meeting at K} | \text{ not met before } K) = \frac{\mathbf{1}_n^T P_{meet} P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]}{\mathbf{1}^T P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]} = \frac{c^T v_{K-1}}{1^T v_{K-1}}$$

$$\min_{\mathrm{v}} \frac{c^T v}{1^T v} = \min_{v \in \Delta} c^T v = \min_i c_i$$

$$\boxed{\frac{\mathbf{1}_n^T P_{meet} P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]}{\mathbf{1}_n^T P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]} \geq \frac{1}{1+\lambda_2} [\mathbf{1}_n^T \quad \mathbf{0}^{\boldsymbol{T}}] \left(P_{\mathcal{G}^2}\right)^K \rho^2[0]}$$

$$\frac{\mathbf{1}^T P_{\neg meet}^K \rho_{\neg meet}^2[0]}{\mathbf{1}^T P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]} \leq \omega \leq \frac{1+\lambda_2}{2} \leq 1$$

$$[\mathbf{1}_n^T \quad \mathbf{0}^{\boldsymbol{T}}] \left(P_{\mathcal{G}^2}\right)^K \rho^2[0] = [\mathbf{1}_n^T A \quad \mathbf{1}_n^T P_{meet}] \left(P_{\mathcal{G}^2}\right)^{K-1} \begin{bmatrix} \rho_{meet}^2 \\ \rho_{\neg meet}^2 \end{bmatrix}$$

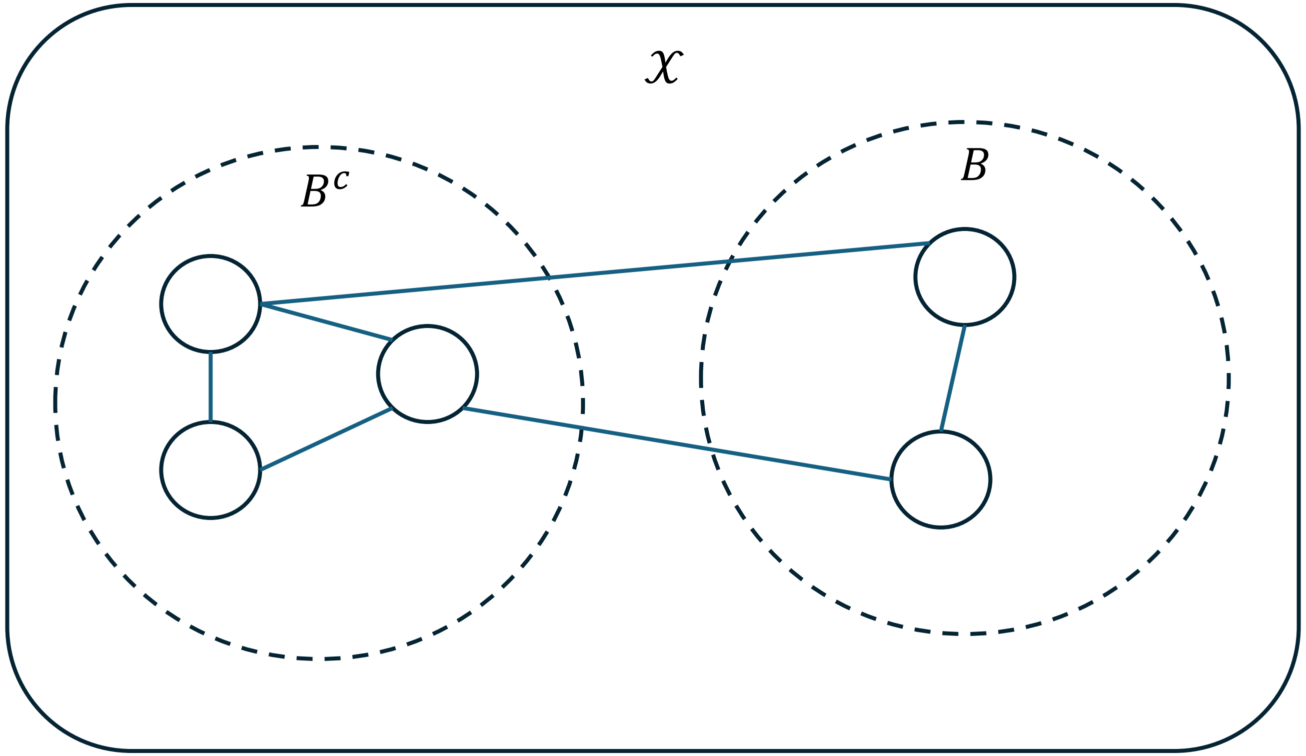
$$\frac{\mathbf{1}^T P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0] - \mathbf{1}_n^T P_{meet} P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]}{\mathbf{1}^T P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]} \leq \frac{1+\lambda_2}{2}$$

$$\frac{\mathbf{1}_n^T P_{meet} P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]}{\mathbf{1}^T P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]} \geq 1 - \frac{1+\lambda_2}{2}$$

$$\frac{\mathbf{1}_n^T P_{meet} P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]}{\mathbf{1}^T P_{\neg meet}^{K-1} \rho_{\neg meet}^2[0]} \geq \frac{1-\lambda_2}{2}$$

Metastability

Metastability



Flux ratio

$$Q_{p,P} = \frac{\sum_{i,j\in B} P_{i,j} p_i}{\sum_{i\in B} p_i}$$

Probability of staying in B given already in B

Metastability

$$\omega_{p,P} = \max_{B\in\mathcal{X},P(B)\leq\frac{1}{2}} \frac{\sum_{i,j\in B} P_{i,j} p_i}{\sum_{i\in B} p_i}$$

Maximum flux ratio across all possible subset

$$2\omega_{p,P} - 1 \leq \lambda_{P,2}$$

$$\frac{1}{\omega_{p,P}} \geq \frac{2}{\lambda_2 + 1}$$

Conductance

$$\psi_{p,P} = \min_{B\in\mathcal{X},P(B)\leq\frac{1}{2}} \frac{\sum_{i,\in \mathrm{B}\ \ j\in B^c} P_{i,j} p_i}{\sum_{i\in B} p_i}$$

Minimum flux ratio from B to B^c across all possible B

$$\frac{P(\text{informed at } k+1)P(\text{uninformed at k})}{P(\text{uninformed at } k+1\ )} \geq \frac{2}{1+\lambda_2} P(\text{informed at } k+1)$$

Conditional informed probability

$$Q(\text{uninformed at k}) \leq \max_B Q(B) = \omega$$

$$\frac{1}{Q(\text{uninformed at k})} \geq \frac{1}{\omega} \geq \frac{2}{1+\lambda_2}$$

$$\frac{P(\text{informed at } k+1)}{Q(\text{uninformed at k})} \geq \frac{2}{1+\lambda_2} P(\text{informed at } k+1)$$

$$\frac{P(\text{informed at } k+1)P(\text{uninformed at k})}{\sum_{i,j} p_i P_{i,j}} \geq \frac{2}{1+\lambda_2} P(\text{informed at } k+1)$$

$$\frac{P(\text{informed at } k+1)P(\text{uninformed at k})}{P(\text{uninformed at } k+1 \text{ \& uninformed at } k)} \geq \frac{2}{1+\lambda_2} P(\text{informed at } k+1)$$

$$\frac{P(\text{informed at } k+1)P(\text{uninformed at k})}{P(\text{uninformed at } k) - P(\text{informed at } k+1 \text{ \& uninformed at } k)} \geq \frac{2}{1+\lambda_2} P(\text{informed at } k+1)$$

$$\frac{P(\text{uninformed at } k) - P(\text{informed at } k+1 \text{ \& uninformed at } k)}{P(\text{informed at } k+1)P(\text{uninformed at k})} \leq \frac{1+\lambda_2}{2P(\text{informed at } k+1)}$$

objective

$$\frac{P(\text{informed at } k{+}1 \text{ \& uninformed at } k)}{P(\text{uninformed at k})} \geq \frac{1}{1+\lambda_2} P(\text{informed at } k+1)$$

B →all possible uninformed state at k

B^c →all possible informed state at k

objective

$$\frac{P(\text{informed at } k{+}1 \text{ \& uninformed at } k)}{P(\text{uninformed at k})} \geq \frac{1}{1+\lambda_2} P(\text{informed at } k+1) \geq \frac{1-\lambda_2}{2}$$

$$1 - \frac{P(\text{informed at } k+1 \text{ \& uninformed at } k)}{P(\text{uninformed at k})} \leq \frac{1+\lambda_2}{2}$$

$$1 - \frac{1+\lambda_2}{2} \leq \frac{P(\text{informed at } k+1 \text{ \& uninformed at } k)}{P(\text{uninformed at k})}$$

$$\frac{1-\lambda_2}{2} \leq \frac{P(\text{informed at } k+1 \text{ \& uninformed at } k)}{P(\text{uninformed at k})}$$

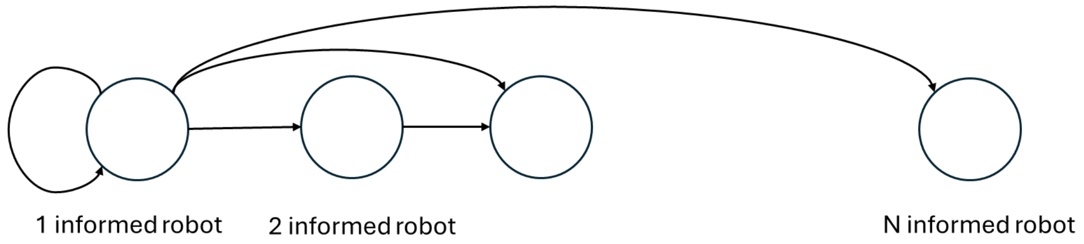
$$\frac{(1-\lambda_2)P(\text{informed at k+1})}{2P(\text{informed at k+1})} \leq \frac{P(\text{informed at } k+1 \text{ \& uninformed at } k)}{P(\text{uninformed at k})}$$

Full Absorbing Chain Method

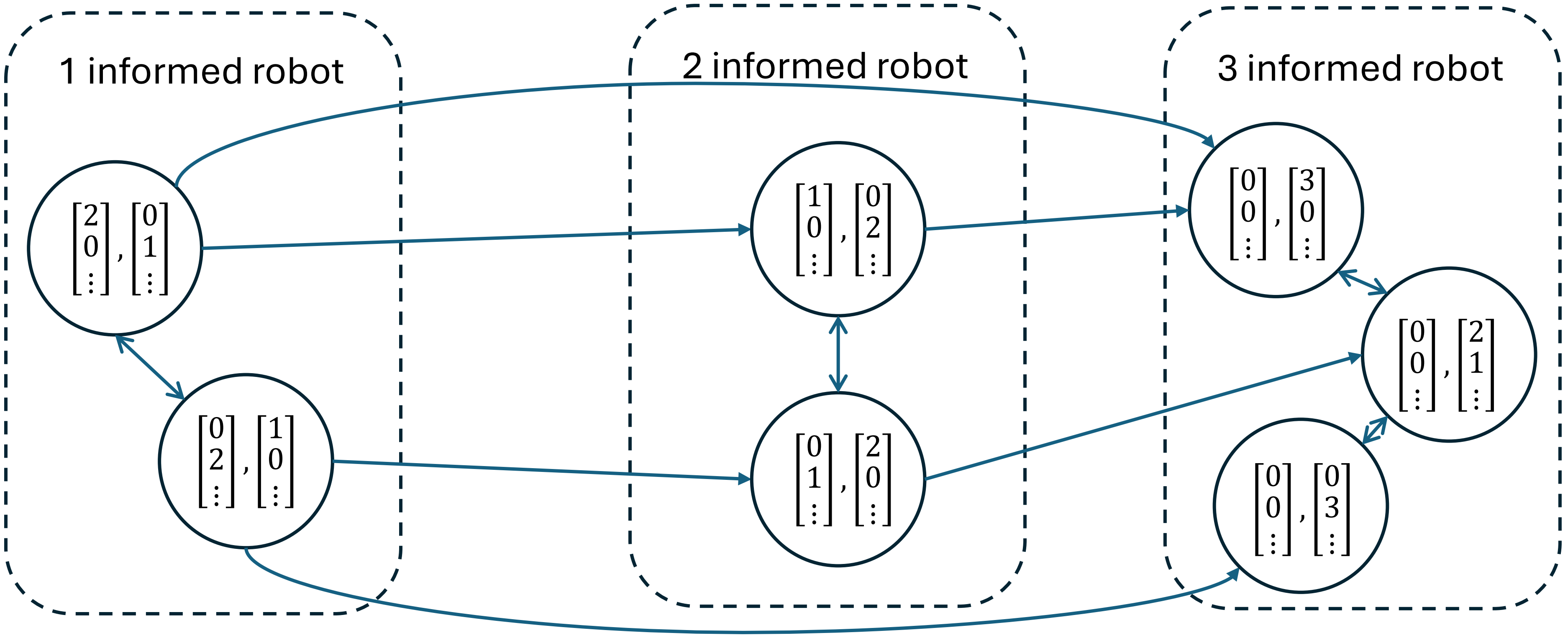
- Mean field method requires λ_2 correction
    - Because ignored graph structure
    - Informed probability is a matrix not a scalar
        - Need to estimate where the robots are informed

Region-aware absorbing chain

Expand nodes to sub-graphs



2 uninformed robot in region 1,
1 informed robot in region 2



N=3, n=2

$$p_{11}^2 p_{22} + 2p_{11}p_{21}p_{22} + p_{11}^2 p_{12} + 2p_{12}p_{11}p_{21} + p_{12}p_{21}^2 + p_{21}^2 p_{22} = 1$$

$$I = 1 \qquad\qquad I = 2$$

$$\begin{bmatrix}2\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix} \quad \begin{bmatrix}0\\2\end{bmatrix},\begin{bmatrix}1\\0\end{bmatrix} \quad \begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\2\end{bmatrix} \quad \begin{bmatrix}0\\1\end{bmatrix},\begin{bmatrix}2\\0\end{bmatrix}$$

$$P = \left[\begin{array}{cc|cc|c} p_{11}^2 p_{22} & p_{12}^2 p_{21} & 0 & 0 & 0 \\ p_{12}p_{21}^2 & p_{11}p_{22}^2 & 0 & 0 & 0 \\ \hline 2p_{11}p_{21}p_{22} & 2p_{21}p_{22}p_{12} & p_{11}p_{22}^2 & p_{12}p_{21}^2 & 0 \\ 2p_{12}p_{11}p_{21} & 2p_{22}p_{12}p_{11} & p_{12}^2 p_{21} & p_{11}^2 p_{22} & 0 \\ \hline p_{11}^2 p_{12} + p_{21}^2 p_{22} & p_{22}^2 p_{21} + p_{12}^2 p_{11} & 1 - p_{11}p_{22}^2 - p_{12}^2 p_{21} & 1 - p_{12}p_{21}^2 + p_{21}p_{22}^2 & 1 \end{array}\right]$$

$$= \begin{bmatrix} Q_{11} & 0 & 0 \\ Q_{21} & Q_{22} & 0 \\ q_{31}^T & q_{32}^T & 1 \end{bmatrix}$$

$$\Pr(1 \text{ informed at K}) = 1^T Q_{11}^K \begin{bmatrix}\rho_1^2\rho_2 \\ \rho_1\rho_2^2\end{bmatrix}$$

$$\Pr(2 \text{ informed exactly at K}) = 1^T Q_{21} Q_{11}^{K-1} \begin{bmatrix}\rho_1^2\rho_2 \\ \rho_1\rho_2^2\end{bmatrix}$$

$$\Pr(3 \text{ informed exactly at K}) = \begin{bmatrix} q_{31}^T & q_{32}^T \end{bmatrix} \begin{bmatrix} Q_{11} & 0 \\ Q_{21} & Q_{22} \end{bmatrix}^{K-1} \begin{bmatrix}\rho_1^2\rho_2 \\ \rho_1\rho_2^2 \\ \rho_1\rho_2^2 \\ \rho_1^2\rho_2\end{bmatrix}$$

$$\Pr(\text{informed at } k) \approx \frac{1}{1+\lambda_2} \sum_{i=0} \rho_i[k](1 - \left(1 - \frac{I_i[k]}{\sum_j I_j}\right)^{\sum_j I_j})$$

$$Q_{rn} \approx p_{rn} = C_r^{N-n} (\Pr(\text{informed at } k))^r (1 - \Pr(\text{informed at } k))^{N-n-r}$$

$$\lambda(P) = \bigcup \lambda(Q_{ii})$$

How many way to arrange i informed robots out of N robots in n regions, such that no uninformed robots get informed

$$\dim(Q_{ii}) = C_i^{n+i-1}C$$